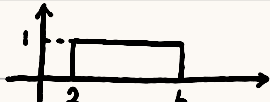
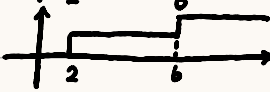


HW1

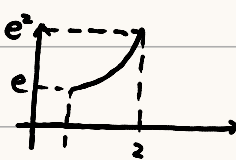
P1. (a)



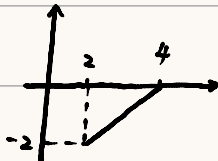
(b)



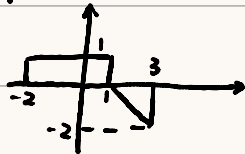
(c)



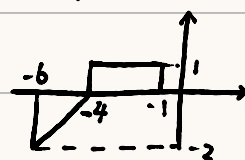
(d)



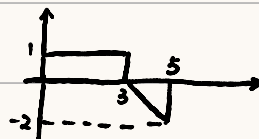
P2. (a)



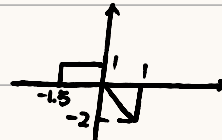
(b)



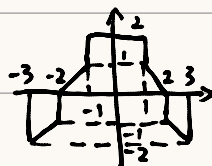
(c)



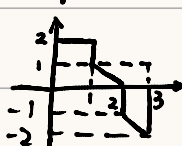
(d)



(e) $x(t) + x(-t)$:



$[x(t) + x(-t)] u(t)$



P3. (a) periodic $T = \frac{\pi}{2}$

(b) $e^{j(\pi t - 1)} = \cos(\pi t - 1) + j \sin(\pi t - 1)$

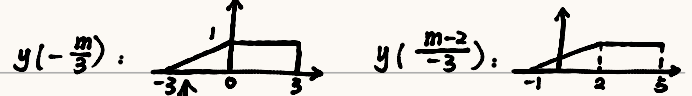
$T = \frac{2\pi}{\pi} = 2$

(c) $x(t) = \frac{1}{2} + \frac{1}{2} \cos(4t - \frac{2\pi}{3})$ $T = \frac{\pi}{2}$

(d) $x(t) = \frac{1}{2} [\cos(4\pi t) u(t) + \cos(4\pi t) u(-t)]$
 $= \cos 4\pi t$ $T = \frac{1}{2}$

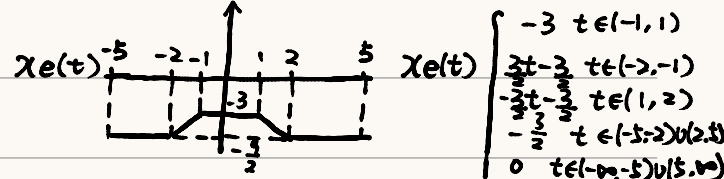
P4. $y(t) = -\frac{1}{9} x(-3t+2)$ $-3t+2=m$ $t = \frac{m-2}{-3}$

$x(m) = -9y(\frac{m-2}{-3})$ $x(t) = -9y(\frac{m-2}{-3})$

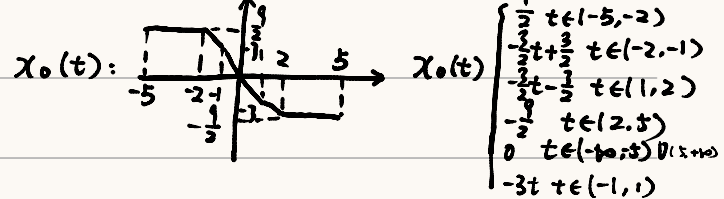


(a) $x(t) = \begin{cases} -3t-3 & t \in (-1, 2) \\ -9 & t \in (2, 5) \\ 0 & t \in (-\infty, -1) \cup (5, \infty) \end{cases}$

(b) $x_e(t) = \frac{x(t) + x(-t)}{2}$



(c) $x_o(t) = \frac{x(t) - x(-t)}{2}$



P5. $P = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t) dt$

(a) $x_1^2(t) = 25 + 100 \cos(100t + \frac{\pi}{3}) + 100 \cos^2(100t + \frac{\pi}{3})$

$P = 25 + 100 \times \frac{1}{2} = 75$

(b) $x_2^2(t) = 100 \cos^2(100t + \frac{\pi}{3}) + 256 \sin^2(150t + \frac{\pi}{3}) + 320 \cos(100t + \frac{\pi}{3}) \sin(150t + \frac{\pi}{3})$

$P = 100 \times \frac{1}{2} + 256 \times \frac{1}{2} = 178$

(c) $x_3^2(t) = \cos^2 10t (100 + 40 \sin 3t + 4 \sin^2 3t)$
 $= (\frac{1}{2} + \cos 20t) (100 + 40 \sin 3t + 2 - 2 \cos 6t)$
 $= 50 + 20 \sin 3t + 1 - \cos 6t + 100 \cos 20t$
 $+ 40 \cos 20t \sin 3t + 20 \cos 20t - 2 \cos 20t \cos 6t$
 $= 51$

$$(d) \quad x_4^2(t) = 100 \cos^2 5t \cos^2 10t$$

$$= 100 \left(\frac{1 + \cos 10t}{2} \right) \left(\frac{1 + \cos 20t}{2} \right)$$

$$= 25 (1 + \cos 10t)(1 + \cos 20t)$$

$$= 25 + 25 \cos 10t + 25 \cos 20t + 25 \cos 10t \cos 20t$$

$$P = 25$$

$$(e) \quad x_5(t) = (\cos \alpha t + i \sin \alpha t) \cos(2\pi W t)$$

$$\textcircled{1} \quad V_0 = 0 \quad x_5(t) = \cos \alpha t + i \sin \alpha t$$

$$|x_5(t)|^2 = 1$$

$$\textcircled{2} \quad V_0 \neq 0 \quad |x_5(t)|^2 = |e^{i\alpha t}|^2 \cos^2 2\pi W t$$

$$= \frac{1}{2}$$

$$\begin{cases} \frac{1}{2} & W \neq 0 \\ 1 & W = 0 \end{cases}$$